Name of the student: Aryan   Sch. No. 231130122

MAULANA AZAD NATIONAL INSTITUTE OF TECHNOLOGY, BHOPAL
Department of Electrical Engineering

End Term Examination, Nov. 2024

Course: B.TECH   **Semester: III**   **Branch: Electrical Engg.**

Subject Name: Electromagnetic Fields &Materials   **Subject code: EE-211**

Time: 03 Hrs.   **Max. Marks: 50**

Note: Answer all questions. Assume missing data if necessary.

Q. No.	Questions	Marks	
I.	a) State the Gauss law and derive the relation $\nabla \bullet D = \rho_v$. Where D is the electric flux density and ρ_v is the volume charge density.	05	CO2
	b) Determine Electric field intensity using Gauss law due i) Infinite surface charge having charge density $\rho_S\ C/m^2$	02	CO1
	ii) Infinite line charge having charge density $\rho_l\ C/m$	02	
II.	a) Solve the potential function between the conductors of coaxial cable by Laplace's equation, if V_0 is the potential difference between them. Also determine the expression for the electric field intensity at any point in the system.	05	CO2
	b) Verify that the field $\vec{A} = yz\,\hat{x} + zx\,\hat{y} + xy\,\hat{z}$ is both irrotational and solenoidal	04	CO1
III	a) A homogeneous dielectric ($\varepsilon_r = 2.5$) fills region-1 ($x \le 0$) while region-2 ($x \ge 0$) is free space. If $D_1 = 12\hat{x} - 10\hat{y} + 4\hat{z}$ nC/m^2. Find D_2 & E_2.	04	CO1
	b) The point charge -1nc, 4nc and 3nc are located at (0, 0, 0), (0 0 1) and (1 0 0) m respectively. Find the energy in the system.	04	CO2
IV	a) Derive the modified Maxwell's second equation based on Ampere's law	04	CO3
	b) Region-1 defined by $Z < 0$ is material for which $\mu_r = 5$, similarly region-2 defined by $Z > 0$ is material for which $\mu_r = 3$, if it is known that no surface current is flowing and $H_2 = 10\hat{x} + 20\hat{z}$. Find B_1, B_2 and H_1.	04	CO3

V.	a)	Drive the generalized equation of energy stored in magnetic fields.	04	CO3
	b)	For a uniform plane wave determine the ratio of magnitude of Electric field to magnetic field in air medium.	04	CO4
VI	a)	Explain the thermal classification of insulating materials and discuss their application in electrical machines.	04	CO5
	b)	Discuss the properties and applications of soft and hard magnetic materials.	04	CO5

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